

Validation Requirement Derivation Module (VRDM)

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· ORA™ · AGA™ · AIG® · CLIA® · MGIA™ · ASGA™ · RIS™
AI-Readable: Yes
Authority: OOF®
Protection: MIP® — Methodological Intellectual Property
Canonical Language: English (UCL™)

Governed Space

Validation Requirement Derivation

Canonical Definition

Validation Requirement Derivation Module (VRDM) defines the governance framework for identifying, deriving, qualifying, documenting, and maintaining the requirements from which formal validation criteria may legitimately be established.

It governs the transformation of relevant operational, technical, scientific, legal, regulatory, contractual, integrity, safety, risk, governance, and purpose-specific requirements into traceable

Validation Requirements suitable for subsequent criterion definition.

Operational Role

VRDM governs the requirement-derivation layer of Validation Criteria governance.

Its role is to establish **where validation criteria legitimately come from** before those criteria are formally defined.

The module creates a governed bridge between:

Validation Mandate → Validation Requirements → Validation Criteria

VRDM prevents criteria from appearing without an identifiable methodological basis.

Module Operational Space

VRDM governs:

- validation requirement identification,
 - requirement derivation,
 - requirement qualification,
 - requirement relevance,
 - requirement origin,
 - requirement authority,
 - requirement applicability,
 - requirement interpretation,
 - requirement dependencies,
 - requirement conflicts,
 - requirement completeness,
 - requirement changes,
 - requirement traceability,
 - and requirement-to-criterion transition.
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Module Function

VRDM applies after the Validation Object and Validation Mandate have been sufficiently established.

Its function is to prevent:

- arbitrary criteria creation,
 - criteria without identifiable requirements,
 - irrelevant requirements entering validation,
 - mandatory requirements being overlooked,
 - recommendations being mistaken for obligations,
 - requirements being detached from their source or authority,
 - assumptions being represented as formal requirements,
 - conflicting requirements being silently combined,
 - and validation criteria being constructed without traceable justification.
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Validation Requirement Sources

Validation Requirements may originate from multiple legitimate sources.

These may include:

Validation Purpose Requirements

Conditions arising directly from what the validation is intended to establish.

Operational Requirements

Conditions necessary for the Validation Object to perform its intended operational role.

Technical Requirements

Relevant specifications, interfaces, tolerances, configurations, performance conditions, or technical constraints.

Scientific Requirements

Conditions supported by applicable scientific knowledge, validated models, established research, or domain methodology.

Safety Requirements

Conditions necessary to govern unacceptable harm, hazardous states, or safety-critical operation.

Integrity Requirements

Conditions necessary to establish or preserve relevant integrity properties.

Risk Requirements

Conditions arising from identified risks, risk thresholds, exposure conditions, or required risk treatments.

Legal and Regulatory Requirements

Applicable statutory, regulatory, supervisory, jurisdictional, or formally mandated conditions.

Contractual Requirements

Conditions established through applicable contractual obligations or service commitments.

Governance Requirements

Conditions originating from approved governance architectures, standards, policies, authorities, or institutional requirements.

Domain Requirements

Sector-specific conditions applicable to healthcare, finance, critical infrastructure, industrial operations, autonomous systems, AI, or another relevant domain.

Requirement Derivation

Not every available requirement automatically becomes a Validation Requirement.

VRDM requires a derivation process capable of establishing why the requirement is relevant to the governed validation.

The basic relationship is:

Requirement Source → Requirement Interpretation → Validation Relevance → Applicability → Validation Requirement

This prevents large collections of external requirements from being imported into validation without determining their actual relationship to the Validation Purpose and Scope.

Requirement Qualification

A candidate requirement should be qualified before being used to create validation criteria.

Qualification may determine:

- whether the requirement is applicable,
- whether it is mandatory or discretionary,
- whether it is current,
- whether its authority is identifiable,
- whether its meaning is sufficiently clear,
- whether it applies to the Validation Object,
- whether it applies to the relevant use or environment,
- whether it materially affects the Validation Purpose,
- and whether it can legitimately support criterion formation.

A requirement that cannot yet be sufficiently qualified should not silently acquire the same status as a confirmed Validation Requirement.

Requirement Authority

Different requirements may carry different forms of authority.

A requirement may originate from:

- law,
 - regulation,
 - contractual obligation,
 - technical specification,
 - scientific reference,
 - organizational governance,
 - external standard,
 - risk determination,
 - operational necessity,
 - or another legitimate source.
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VRDM preserves the origin and nature of that authority.

It does not automatically assume that all requirement sources have equal governance weight.

Requirement Relevance

A requirement is relevant when it has a defensible relationship to the approved validation.

Relevance should be established against:

- the Validation Object,
- Validation Purpose,
- Validation Scope,
- Validation Depth,
- Validation Applicability,
- and known Validation Limitations.

A valid requirement in one context may be irrelevant to another validation.

VRDM therefore prevents general validity from being confused with validation-specific relevance.

Requirement Applicability

Even a relevant requirement may apply only under defined conditions.

Applicability may depend upon:

- jurisdiction,
- system configuration,
- operational environment,
- intended use,
- user population,
- lifecycle stage,
- deployment mode,
- risk level,
- dependency state,
- or another material condition.

Conditional applicability must remain explicit when the Validation Requirement is transferred into criterion governance.

Requirement Interpretation

Some source requirements may not be expressed in a form directly usable for validation.

VRDM allows them to be interpreted into validation-relevant requirements while preserving traceability to the original source.

Interpretation must not materially change the meaning of the originating requirement merely to make validation easier.

Where interpretation is uncertain or contested, that uncertainty should remain visible.

Requirement Dependencies

Validation Requirements may depend upon one another.

For example:

Requirement A may apply only when **Requirement B** is satisfied.

A higher-level requirement may require multiple subordinate conditions.

One requirement may activate another.

VRDM identifies material dependency relationships before formal criteria are constructed.

Requirement Conflict

Requirements may conflict because of:

- different authorities,
- different jurisdictions,
- incompatible operational objectives,
- competing safety and performance conditions,
- inconsistent technical specifications,
- changing scientific knowledge,
- or differences between governance frameworks.

VRDM requires material conflicts to be identified rather than silently resolved during criterion creation.

Where resolution belongs to another governance authority, the conflict should be escalated or preserved as unresolved rather than arbitrarily decided within validation.

Requirement Completeness

VRDM assesses whether all material requirement categories necessary for the approved Validation Mandate have been considered.

Requirement completeness does not mean collecting every possible requirement.

It means determining whether a material source of validation obligations or conditions has been omitted.

A validation process should not appear complete merely because the requirements that were easiest to identify were included.

Minimum Implementation Framework

1. Identify Requirement Sources

Determine which operational, technical, scientific, safety, integrity, risk, legal, regulatory, contractual, governance, and domain sources may contain requirements relevant to validation.

2. Derive Candidate Requirements

Translate relevant source conditions into identifiable candidate Validation Requirements.

3. Qualify Requirements

Determine:

- relevance,
- applicability,
- authority,
- currency,
- clarity,
- and relationship to the Validation Mandate.

4. Identify Dependencies and Conflicts

Determine whether requirements:

- depend upon other requirements,
- activate other requirements,
- overlap,
- contradict one another,
- or require external governance resolution.

5. Establish Validation Requirements

Only sufficiently qualified requirements should become governed Validation Requirements capable of supporting criterion definition.

6. Preserve Requirement Traceability

Maintain:

- requirement identifier,
- requirement source,

- originating condition,
- derivation logic,
- authority basis,
- applicability,
- qualification status,
- dependencies,
- conflicts,
- version,
- changes,
- and relationship to resulting Validation Criteria.

Requirement Change Governance

Validation Requirements may change during or after validation.

Changes may arise from:

- regulatory amendments,
- new scientific knowledge,
- revised technical specifications,
- changed operational conditions,
- changed risk,
- changed intended use,
- new governance requirements,
- contractual modification,
- or changes to the Validation Object itself.

A material requirement change must trigger assessment of its effect upon existing Validation Criteria and downstream validation activity.

Requirement-to-Criterion Transition

VRDM ends where formal criterion definition begins.

A Validation Requirement identifies **what condition must be represented within validation governance.**

The resulting Validation Criterion determines **how that requirement is expressed as an explicit condition against which the Validation Object can later be evaluated.**

Therefore:

Validation Requirement ≠ Validation Criterion

The governed transition is:

Qualified Validation Requirement → Criterion Definition

This separation prevents source requirements from being copied into

validation without the methodological transformation necessary to make them evaluable.

Use Case 1 — AI System Deployment

Scenario

An organization intends to validate an AI system for deployment within a regulated operational environment.

Multiple technical specifications, internal policies, risk controls, regulatory requirements, and operational expectations exist.

Application

VRDM identifies the relevant requirement sources and determines which conditions legitimately apply to the Validation Object, intended use, environment, and Validation Purpose.

Requirements are qualified before they are transferred into criterion definition.

Result

The Validation Criteria Framework is built from traceable requirements rather than an arbitrary checklist assembled by the validation team.

Use Case 2 — Critical Infrastructure Validation

Scenario

A system supporting critical infrastructure is subject to operational, safety, technical, contractual, and regulatory requirements.

Some requirements overlap while others establish different thresholds.

Application

VRDM derives and qualifies the relevant Validation Requirements, identifies dependencies and conflicts, and preserves their authority and origin.

Result

Subsequent validation criteria can be constructed from an explicit and traceable requirement foundation without silently merging materially different obligations.

Canonical Closing Statement

Validation Requirement Derivation Module establishes the canonical governance framework for determining where validation criteria legitimately originate. By governing requirement identification, derivation, qualification, relevance, authority, applicability, interpretation, dependencies, conflicts, completeness, change, and traceability, VRDM ensures that every material Validation Criterion can be connected to an identifiable and justified Validation Requirement rather than emerging from arbitrary expectation or post-hoc judgment.

Parent Resources

[→ VALIDOS® Validation Governance Architecture — Validation Governance Layer](#)

[→ VALIDOS® Architecture Map](#)

[→ VALIDOS® Complete Standards & Modules Index](#)

Internal Module Flow

[→ Next module: Validation Criterion Formulation Module \(VCFM\)](#)

Related Documents

[→ Validation Criteria Standard](#)

[→ About Validation Criteria Standard](#)

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Canonical Source

<https://originopenfoundation.org/>